

MENTIS · Procurement Intelligence Brief

Substance: insulin glargine · **Normalized:** 1.5 ML insulin glargine 300 UNT/ML Pen Injector [Toujeo]

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Executive Summary

Insulin glargine is a long-acting basal insulin analogue crucial for diabetes management. Primarily, it supports patients requiring stable, long-term glycemic control, with Toujeo (300 U/mL) offering an ultra-concentrated formulation [PubMed](#). The market shows steady growth, with a projected CAGR of 6.6%, fueled by rising diabetes prevalence and biosimilar competition [source](#). Regulatory pathways, such as Semglee's FDA approval as an interchangeable biosimilar, have increased affordability and access [ProAct Clinical Corner](#). Sanofi dominates production through Toujeo, but biosimilar manufacturers like Biocon Biologics and Gan & Lee are gaining traction [source](#). Supply constraints and post-patent pricing pressures highlight the importance of strategic procurement and sourcing efficiency [source](#).

Product Profile

Insulin glargine, a long-acting basal insulin analogue, has been integral to diabetes management since its clinical introduction in 2000 [PubMed](#). It replaced intermediate-acting insulins like neutral protamine Hagedorn (NPH), offering once-daily administration to better control basal metabolism. Among the formulations available, Toujeo (1.5 ML pen injector, 300 UNT/ML) is particularly notable for providing an ultra-long-acting insulin alternative.

The development of biosimilars has further expanded the insulin glargine market. Semglee, approved by the U.S. Food and Drug Administration (FDA) in July 2021, represents the first interchangeable biosimilar to Lantus—the reference insulin glargine product [PubMed](#). Such biosimilar approvals promote competitive pricing and broaden patient accessibility, amid rising demand for advanced insulin therapies.

Future innovation in basal insulin analogs is underway, addressing limitations such as impurity formation identified in existing formulations. For instance, insulin lisargine introduces a glycine substitution at A21 to mitigate these issues, improving safety profiles compared to insulin glargine [PubMed](#). These advancements reflect ongoing efforts to enhance efficacy and tolerability in long-term diabetes care.

Continuous glucose monitoring (CGM) systems are increasingly used alongside insulin therapy, especially for Type 2 diabetes patients requiring intensive glycemic management. Such integrations underline the complex, evolving ecosystem supporting insulin glargine utilization [PubMed](#).

Clinical Use & Mechanism

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Market Size & Demand Drivers

The global market demand for insulin glargine, particularly the 300 unit/mL formulation found in Toujeo, is experiencing steady growth, driven by rising diabetes prevalence and advancements in therapeutic delivery systems. The market is projected to grow at a Compound Annual Growth Rate (CAGR) of 6.6%, reaching an estimated value of USD 5.81 billion by 2026, as noted in recent analyses [here](#). Regionally, Asia-Pacific is the fastest-growing market, with a CAGR of 9.05%, owing to policy interventions and reduced drug prices in countries like China [source](#).

Toujeo, a key product in this space, has seen considerable market performance. In 2025, its sales increased by 6.5%, with Europe and the Rest of the World contributing significantly to this growth, while U.S. sales faced a slight decline due to subsiding windfall effects [details](#). However, the market faces headwinds from the growing competition posed by biosimilars such as Semglee, particularly in developed markets like Japan [read more](#).

Emerging economies are also shaping market dynamics. For instance, India's recent regulatory approvals for needle-free delivery and expansion of domestic manufacturing have enhanced market accessibility [insight](#). In contrast, North America's insulin pen market, including glargine, is set to grow at a robust CAGR of 10.1% during 2025–2034, reflecting heightened demand for innovative insulin delivery systems [context](#). These trends collectively underscore the sustained, albeit competitive, global demand for insulin glargine products.

Top Manufacturers & Suppliers

Insulin glargine, a long-acting analog of insulin, is produced by several key manufacturers, with Sanofi leading production through its proprietary product Toujeo. Sanofi maintains a majority market share, bolstered by its established product portfolio including Lantus and Basaglar. From 2017 to 2018, Toujeo accounted for approximately 32.3% of market offsets, second only to Lantus, which held 60.3% during the same period [source](#).

Emerging competitors like Biocon Biologics and Gan & Lee are gaining traction by securing hospital tenders and interchangeability approvals, presenting biosimilar alternatives to originator brands. The discontinuation of Biocon/Mylan's Semglee biosimilar in January 2025 increased demand for Toujeo and other insulin glargine products [source](#). Meanwhile, Sunshine Lake Pharma and Yifan Pharmaceutical also engage in production, particularly serving price-sensitive markets via collaborations and commercial-scale supply chains [source](#) [source](#).

Geographically, Novo Nordisk continues to expand its footprint in emerging markets, including direct investments in Algeria and Kenya, while fostering growth in Latin America at 8.1% CAGR. North America and Europe exhibit mature oligopolistic competition, whereas Asia and Africa remain fragmented [source](#). As diabetes prevalence rises globally, manufacturers increasingly target

affordability and education to sustain market competitiveness, reflecting both consolidation and diversification strategies among industry leaders.

Regulatory & Compliance

Regulatory developments around insulin glargine, particularly in its biosimilar forms, underscore its evolving landscape within the pharmaceutical industry. **Toujeo** (1.5 mL insulin glargine 300 units/mL pen injector) was approved with defined specifications, supplying this formulation in prefilled packages designed for clinical convenience [FDA labeling](#). Its approval highlights the regulatory pathway for a unique high-strength insulin glargine product.

Equally notable is the entry of **Semglee®** as a biosimilar competitor. Initially approved under the 505(b)(2) NDA pathway in June 2020, Semglee was prescribed similarly to Lantus but lacked full biosimilar or interchangeability status at the time [Drugs.com history](#). This changed on July 28, 2021, when Semglee became the FDA's first interchangeable biosimilar insulin product, allowing substitution with Lantus without prescriber approval [ProAct Clinical Corner](#).

Both Toujeo and Semglee address distinct market needs: Toujeo targets patients requiring concentrated insulin formulations, while Semglee provides a cost-effective, interchangeable biosimilar option. The regulatory milestone achieved by Semglee signals significant potential for cost savings and broadened access within the insulin market, which continues to be marked by innovation and competitive dynamics.

Sourcing & Pricing Channels

Toujeo, a high-dose insulin glargine (300 IU/mL) produced by Sanofi, is widely used for diabetes management globally. It is commonly prescribed for individuals requiring long-acting insulin to control blood sugar levels, particularly in Europe where demand in countries like Germany, France, and the U.K. remains robust [here](#). However, as of early 2026, intermittent supply constraints have been reported, though Toujeo has not been officially listed in shortage by the FDA [here](#).

Pricing dynamics for insulin glargine have shifted significantly due to the patent expiration of older formulations like Lantus. Biosimilars, such as Semglee and Abasaglar, have entered markets with price reductions of 15-30% compared to branded options, pressuring Toujeo's competitive positioning in some areas [here](#). As of Q4 2025, Toujeo's Wholesale Acquisition Cost (WAC) stood at \$95 per unit, which is higher than comparable formulations like Basaglar (\$65) [here](#).

In developing markets, pricing remains a critical factor. For example, a cost-utility analysis suggested biosimilar insulin glargine could achieve cost-effectiveness at price thresholds significantly lower than its branded counterparts, highlighting opportunities for less expensive alternatives in resource-constrained settings [here](#). These pricing disparities emphasize the importance of effective procurement strategies, especially in public-sector distribution networks.

Risks & Alternatives

Insulin glargine, particularly in its concentrated form as Toujeo (300 U/mL), is a cornerstone therapy for diabetes management, offering prolonged blood sugar control. A key risk associated with insulin

glargine is hypoglycemia. Contributing factors include overdosing, missed meals, or changes in injection routines. Severe cases of hypoglycemia can lead to unconsciousness, especially if not addressed promptly [source](#). Pediatric studies indicate that children with type 1 diabetes are at a higher risk of severe symptomatic hypoglycemia compared to adult patients [source](#).

Alternatives to Toujeo include biosimilars such as Basaglar (Eli Lilly), which shares the same active ingredient at identical concentrations. Basaglar can often be substituted on a 1:1 dose basis with a physician's approval. Generic insulin glargine formulations are also available and therapeutically equivalent [source](#). Tresiba (insulin degludec) is another option, offering advantages such as extended duration of action and potentially improved glycemic control for some patients. However, differences in pharmacokinetic profiles may make direct comparisons complex [source](#).

For older adults, evidence suggests that Toujeo significantly reduces hypoglycemia risks compared to other basal insulins, including insulin detemir and insulin glargine 100 U/mL. The DELIVER 31 study observed a 57% reduction in hypoglycemic events for patients aged 65 and older switching to Toujeo [source](#). These findings may guide clinicians in tailoring treatments to patient-specific risks and needs.

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